



ACCENTURE INNOVATION CHALLENGE 2026 • OFFICIAL TECHNICAL README

ControlPlane.ai

Model-Agnostic Real-Time AI Oversight, Guardrail & Quality Control Tower

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1. Executive Overview & Problem Statement

As organizations rush Generative AI into customer support, coding, executive decision-making, and internal workflows, one critical question stops being theoretical: **What happens the moment an AI gives an answer that sounds right — but isn't?**

AI systems can be confidently wrong, generate misleading answers with authoritative tone, quietly burn compute through unnecessary tokens, and expose organizations to sensitive-data leakage without a single traditional infrastructure alarm going off.

\$67.4 Billion

HALLUCINATION BUSINESS LOSS

Estimated global enterprise cost of AI hallucinations in 2024 (AllAboutAI / Korra Report).

97%

SECURITY ACCESS FAILURES

Of organizations reporting AI security breaches lacked real-time inline access guardrails (IBM 2025 Study).

63%

LACK FORMAL GOVERNANCE

Enterprises deploy LLMs to production without automated quality gates, relying solely on uptime logs (Deloitte).

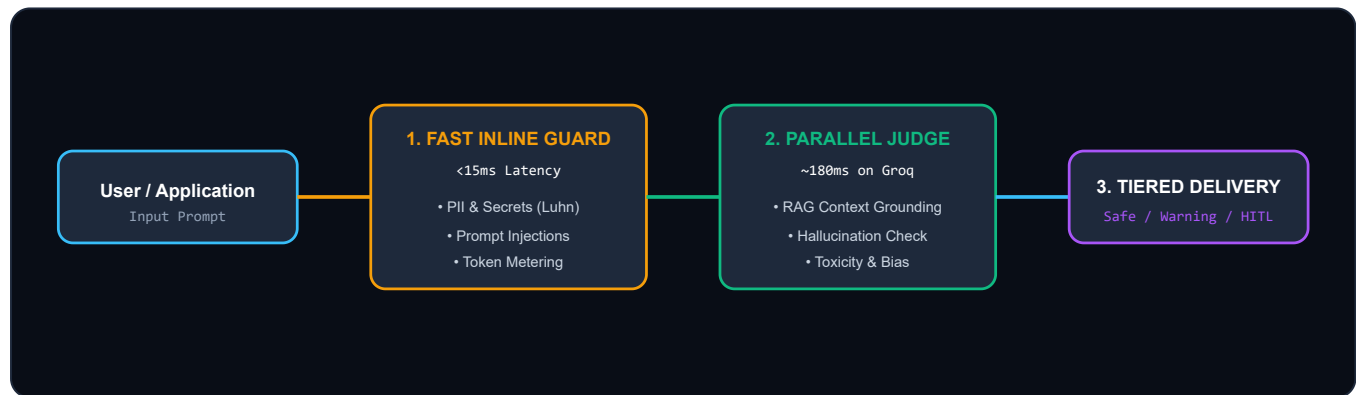
< 15 ms

FAST INLINE SLA

Synchronous PII and injection inspection intercepts attacks without adding user lag.

2. System Architecture: The Dual-Speed Engine

ControlPlane.ai recognizes that **not all guardrails need to run at the same speed**. Fast signals run synchronously inline, while deep judge models verify factual groundedness in parallel.



3. Why We Used These Technologies (Design Rationale)

TECHNOLOGY	LAYER	ENGINEERING RATIONALE & STRATEGIC WHY
Groq Cloud LPUs	Inference Engine	Provides <200ms time-to-first-token inference for 70B and 8B models. Enables live LLM-as-a-Judge evaluation in parallel with zero perceivable delay to the end user.
Meta Llama 3 & Compound Models	Base & Judge LLM	Open-weights, enterprise-safe, with superior reasoning on structured JSON output and factual comparison benchmarks.
FastAPI (Python)	Backend Gateway	Asynchronous native ASGI framework with built-in Pydantic validation, concurrent background task workers, and high-performance WebSockets.
WebSockets (Native)	Live Telemetry	Instant bidirectional push of telemetry logs to the Control Tower UI with sub-millisecond overhead compared to polling.
React 19 + TypeScript + Vite	Frontend Architecture	Strict type safety across the entire telemetry contract, rapid HMR developer experience, and component-level re-rendering optimization.
Tailwind CSS (Dark Obsidian)	UI / UX Design	High-contrast, accessibility-compliant executive dark aesthetic (#07090E) with glowing status indicators.
Recharts	Data Visualization	Declarative, GPU-accelerated SVG charting for OWASP threat distribution and latency histograms.

4. OWASP Top 10 for LLMs Security Mapping

OWASP ID	VULNERABILITY NAME	CONTROLPLANE.AI DEFENSIVE MECHANISM
LLM01	Prompt Injection	Synchronous heuristic and delimiter analysis stops jailbreak attempts in <15ms.
LLM02	Sensitive Info Disclosure	Regex, Luhn verification, and secret scanners strip/block API keys, credit cards, and SSNs.
LLM03	Hallucination / Misinformation	Parallel Judge model scores groundedness against verified reference documents.
LLM04	Data & Model Poisoning	HITL audit trail isolates anomalous model outputs and triggers administrative review.
LLM06	Excessive Agency	Dynamic Tiered gating prevents unauthorized downstream execution of unverified commands.

OWASP ID	VULNERABILITY NAME	CONTROLPLANE.AI DEFENSIVE MECHANISM
LLM10	Unbounded Consumption	Token metering and adaptive sampling depth caps prevent denial-of-wallet compute spikes.

5. 90-Second Pitch & Live Demo Script

- 1. **The Problem (0:00 – 0:30 on Home / Control Tower):** Explain that traditional APMs only watch uptime and latency — they cannot detect when an AI is confidently wrong. In 2024, hallucinations cost enterprises \$67.4B.
- 2. **The Dual-Speed Engine (0:30 – 1:05 on Gateway Lab):**
 - Click **Preset 1 (Credit Card Exfiltration)**: Show Fast Guard intercepting sensitive PII in 1.2ms.
 - Click **Preset 2 (Financial Hallucination)**: Show Parallel Judge catching the fabricated 34.8% growth metric and attaching a Citation Warning.
- 3. **Threat Surge & HITL Audit (1:05 – 1:30 on Control Tower & HITL Queue):**
 - Click **Threat Surge**: Watch the Adaptive Sampling gauge auto-climb from 25% to 85% to shield infrastructure.
 - Open **HITL Queue**: Demonstrate 1-click remediation and export of timestamped compliance packages.